

ARCTIC OR POLAR CLIMATE

POLAR CLIMATE WORLD DISTRIBUTION → TEMPERATURE-PRECIPITATION PROFILE → TUNDRA VEGETATION BELT → ARCTIC-ANTARCTIC ASYMMETRY → ICE-ALBEDO FEEDBACK LOOP → PERMAFROST CARBON FEEDBACK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g3 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00

POLAR CLIMATE WORLD DISTRIBUTION

POLAR CLIMATE WORLD DISTRIBUTION

GREENLAND AND HIGH-LATITUDE HIGHLANDS, PERMANENT SNOW COVER

COASTAL GREENLAND, N. CANADA, ALASKA, EURASIAN ARCTIC SEABOARD

SOUTHERN POLE

CONTINENT UNDER ICE, NO INDIGENOUS POPULATION

NORTHERN POLE

OCEAN BASIN SURROUNDED BY CONTINENTAL LAND MASSES

ICE-CAP → Greenland and high-latitude highlands, permanent snow cover

TUNDRA → coastal Greenland, N. Canada, Alaska, Eurasian Arctic seaboard

SOUTHERN POLE → Antarctica: continent under ice, no indigenous population

NORTHERN POLE → Arctic: ocean basin surrounded by continental land masses

SYSTEM / DEFINITION

- ICE-CAP → Greenland and high-latitude highlands, permanent snow cover
- TUNDRA → coastal Greenland, N. Canada, Alaska, Eurasian Arctic seaboard

PROCESS / MECHANISM

- SOUTHERN POLE → Antarctica: continent under ice, no indigenous population
- NORTHERN POLE → Arctic: ocean basin surrounded by continental land masses

SPATIAL / INDIA PATTERN

- ICE-CAP → Greenland and high-latitude highlands, permanent snow cover
- TUNDRA → coastal Greenland, N. Canada, Alaska, Eurasian Arctic seaboard

HAZARD / LIMIT / RESPONSE

- SOUTHERN POLE → Antarctica: continent under ice, no indigenous population
- NORTHERN POLE → Arctic: ocean basin surrounded by continental land masses

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NORTHERN POLE → Arctic: ocean basin surrounded by continental land masses.

01

STAGE 01

TEMPERATURE-PRECIPITATION PROFILE

TEMPERATURE-PRECIPITATION PROFILE

WARMEST MONTH

SELDOM EXCEEDS ABOUT 10 DEGREES C (JUNE/JULY)

MONTHS ABOVE FREEZING

NO MORE THAN ABOUT FOUR PER YEAR

LOW, MOSTLY SNOW; POLAR CLIMATE

COLD DESERT

WIND REDISTRIBUTING EXISTING SNOW, NOT HEAVY SNOWFALL EVENTS

DEFINITION / COMPONENTS

WARMEST MONTH → seldom exceeds about 10 degrees C (June/July)

MONTHS ABOVE FREEZING → no more than about four per year

TRIGGER / MECHANISM

PRECIPITATION → low, mostly snow; polar climate = cold desert

BLIZZARDS → wind redistributing existing snow, not heavy snowfall events

EFFECT / INTERVENTION

WARMEST MONTH → seldom exceeds about 10 degrees C (June/July)

MONTHS ABOVE FREEZING → no more than about four per year

DEFINITION / COMPONENTS

- WARMEST MONTH → seldom exceeds about 10 degrees C (June/July)
- MONTHS ABOVE FREEZING → no more than about four per year

TRIGGER / MECHANISM

- PRECIPITATION → low, mostly snow; polar climate = cold desert
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EFFECT / INTERVENTION

- WARMEST MONTH → seldom exceeds about 10 degrees C (June/July)
- MONTHS ABOVE FREEZING → no more than about four per year

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BLIZZARDS → wind redistributing existing snow, not heavy snowfall events.

02

STAGE 02

TUNDRA VEGETATION BELT

TUNDRA VEGETATION BELT

PREVENTS DEEP ROOTING, IMPEDES DRAINAGE

SHORT GROWING SEASON

WARMEST MONTH BELOW 10 DEGREES C

STUNTED PLANTS

DWARF WILLOWS, BIRCHES, ALDERS IN SHELTERED SPOTS

LICHEN GROUND COVER

REINDEER MOSS PROVIDES GRAZING FOR REINDEER HERDS

PERMAFROST → prevents deep rooting, impedes drainage

SHORT GROWING SEASON → warmest month below 10 degrees C

STUNTED PLANTS → dwarf willows, birches, alders in sheltered spots

LICHEN GROUND COVER → reindeer moss provides grazing for reindeer herds

DEFINITION / COMPONENTS

- PERMAFROST → prevents deep rooting, impedes drainage
- SHORT GROWING SEASON → warmest month below 10 degrees C

TRIGGER / MECHANISM

- STUNTED PLANTS → dwarf willows, birches, alders in sheltered spots
- LICHEN GROUND COVER → reindeer moss provides grazing for reindeer herds

EFFECT / INTERVENTION

- PERMAFROST → prevents deep rooting, impedes drainage
- SHORT GROWING SEASON → warmest month below 10 degrees C

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LICHEN GROUND COVER → reindeer moss provides grazing for reindeer herds.

03

STAGE 03

ARCTIC-ANTARCTIC ASYMMETRY

ARCTIC-ANTARCTIC ASYMMETRY

OCEAN SURROUNDED BY CONTINENTS; MAINLY FLOATING SEA ICE

CONTINENT SURROUNDED BY OCEAN; VAST GROUNDED ICE SHEET

ARCTIC AMPLIFICATION MUCH FASTER; ANTARCTIC MORE COMPLEX

SOVEREIGN STATES + FORUM

TREATY REGIME

DEFINITION / COMPONENTS

ARCTIC → ocean surrounded by continents; mainly floating sea ice

TRIGGER / MECHANISM

WARMING → Arctic amplification much faster; Antarctic more complex

EFFECT / INTERVENTION

Antarctic: treaty regime

03

STAGE 03

ARCTIC-ANTARCTIC ASYMMETRY

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OCEAN SURROUNDED BY CONTINENTS; MAINLY FLOATING SEA ICE

CONTINENT SURROUNDED BY OCEAN; VAST GROUNDED ICE SHEET

ARCTIC AMPLIFICATION MUCH FASTER; ANTARCTIC MORE COMPLEX

SOVEREIGN STATES + FORUM

TREATY REGIME

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
ARCTIC → ocean surrounded by continents; mainly floating sea ice	WARMING → Arctic amplification much faster; Antarctic more complex	Antarctic: treaty regime
ANTARCTIC → continent surrounded by ocean; vast grounded ice sheet	GOVERNANCE → Arctic: sovereign states + forum	ARCTIC → ocean surrounded by continents; mainly floating sea ice

DEFINITION / COMPONENTS

- ARCTIC → ocean surrounded by continents; mainly floating sea ice
- ANTARCTIC → continent surrounded by ocean; vast grounded ice sheet

TRIGGER / MECHANISM

- WARMING → Arctic amplification much faster; Antarctic more complex
- GOVERNANCE → Arctic: sovereign states + forum

EFFECT / INTERVENTION

- Antarctic: treaty regime
- ARCTIC → ocean surrounded by continents; mainly floating sea ice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LICHEN GROUND COVER → reindeer moss provides grazing for reindeer herds.

04

STAGE 04

ICE-ALBEDO FEEDBACK LOOP

ICE-ALBEDO FEEDBACK LOOP

SEA ICE AND SNOW RETREAT

BRIGHT SURFACE REPLACED BY DARK OCEAN/GROUND

ALBEDO DROPS

FAR MORE SOLAR RADIATION ABSORBED INSTEAD OF REFLECTED

SURFACE WARMING

FURTHER ICE AND SNOW RETREAT (SELF-REINFORCING)

REINFORCING FACTORS

SEA ICE AND SNOW RETREAT → bright surface replaced by dark ocean/ground

ALBEDO DROPS → far more solar radiation absorbed instead of reflected

SURFACE WARMING → further ice and snow retreat (self-reinforcing)

REINFORCING FACTORS → thin polar atmosphere, ocean heat escape, poleward transport

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
SEA ICE AND SNOW RETREAT → bright surface replaced by dark ocean/ground	SURFACE WARMING → further ice and snow retreat (self-reinforcing)	SEA ICE AND SNOW RETREAT → bright surface replaced by dark ocean/ground
ALBEDO DROPS → far more solar radiation absorbed instead of reflected	REINFORCING FACTORS → thin polar atmosphere, ocean heat escape, poleward transport	ALBEDO DROPS → far more solar radiation absorbed instead of reflected

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REINFORCING FACTORS → thin polar atmosphere, ocean heat escape, poleward transport.

05

STAGE 05

PERMAFROST CARBON FEEDBACK

PERMAFROST CARBON FEEDBACK

THAWS FROZEN GROUND CONTAINING ANCIENT ORGANIC MATTER

DRY CONDITIONS

DECOMPOSITION RELEASES CARBON DIOXIDE

WATERLOGGED CONDITIONS

DECOMPOSITION RELEASES METHANE (STRONGER GHG)

RELEASED GHGS CAUSE FURTHER WARMING, FURTHER THAW

WARMING → thaws frozen ground containing ancient organic matter

DRY CONDITIONS → decomposition releases carbon dioxide

WATERLOGGED CONDITIONS → decomposition releases methane (stronger GHG)

FEEDBACK → released GHGs cause further warming, further thaw

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
WARMING → thaws frozen ground containing ancient organic matter	WATERLOGGED CONDITIONS → decomposition releases methane (stronger GHG)	WARMING → thaws frozen ground containing ancient organic matter
DRY CONDITIONS → decomposition releases carbon dioxide	FEEDBACK → released GHGs cause further warming, further thaw	DRY CONDITIONS → decomposition releases carbon dioxide

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FEEDBACK → released GHGs cause further warming, further thaw.

06

STAGE 06

SEA-LEVEL GROUNDED-ICE RULE

SEA-LEVEL GROUNDED-ICE RULE

FLOATING SEA ICE MELTS

SEA LEVEL DOES NOT RISE (ALREADY DISPLACING WATER)

GROUNDED ICE MELTS

MASS TRANSFERRED TO OCEAN, SEA LEVEL RISES

ICE SHELVES LOST

REMOVES BUTTRESSING, ACCELERATES GROUNDED-ICE DISCHARGE

EXAM RULE

FLOATING SEA ICE MELTS → sea level does NOT rise (already displacing water)

GROUNDED ICE MELTS → mass transferred to ocean, sea level RISES

ICE SHELVES LOST → removes buttressing, accelerates grounded-ice discharge

EXAM RULE → always distinguish floating from grounded in any polar answer

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
FLOATING SEA ICE MELTS → sea level does NOT rise (already displacing water)	ICE SHELVES LOST → removes buttressing, accelerates grounded-ice discharge	FLOATING SEA ICE MELTS → sea level does NOT rise (already displacing water)
GROUNDED ICE MELTS → mass transferred to ocean, sea level RISES	EXAM RULE → always distinguish floating from grounded in any polar answer	GROUNDED ICE MELTS → mass transferred to ocean, sea level RISES

06

STAGE 06 SEA-LEVEL GROUNDED-ICE RULE

SEA-LEVEL GROUNDED-ICE RULE FLOATING SEA ICE MELTS SEA LEVEL DOES NOT RISE (ALREADY DISPLACING WATER) GROUNDED ICE MELTS MASS TRANSFERRED TO OCEAN, SEA LEVEL RISES ICE SHELVES LOST REMOVES BUTTRESSING, ACCELERATES GROUNDED-ICE DISCHARGE EXAM RULE

FLOATING SEA ICE MELTS → sea level does NOT rise (already displacing water)

GROUNDED ICE MELTS → mass transferred to ocean, sea level RISES

ICE SHELVES LOST → removes buttressing, accelerates grounded-ice discharge

EXAM RULE → always distinguish floating from grounded in any polar answer

DEFINITION / COMPONENTS

- FLOATING SEA ICE MELTS → sea level does NOT rise (already displacing water)
- GROUNDED ICE MELTS → mass transferred to ocean, sea level RISES

TRIGGER / MECHANISM

- ICE SHELVES LOST → removes buttressing, accelerates grounded-ice discharge
- EXAM RULE → always distinguish floating from grounded in any polar answer

EFFECT / INTERVENTION

- FLOATING SEA ICE MELTS → sea level does NOT rise (already displacing water)
- GROUNDED ICE MELTS → mass transferred to ocean, sea level RISES

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXAM RULE → always distinguish floating from grounded in any polar answer.

07

STAGE 07 ARCTIC SHIPPING CONSTRAINTS

ARCTIC SHIPPING CONSTRAINTS ROUTES SHORTER THAN SUEZ/PANAMA BETWEEN NE ASIA AND NW MOBILE ICE UNPREDICTABLE, HAZARDOUS DESPITE RETREAT TREND RESCUE AND CHARTING THIN CAPACITY, INCOMPLETE SURVEYS NEAR-TERM TRAFFIC

RESOURCE EXPORT AND DESTINATION, NOT TRANSIT CONTAINER

DEFINITION / COMPONENTS

ADVANTAGE → routes shorter than Suez/Panama between NE Asia and NW Europe
MOBILE ICE → unpredictable, hazardous despite retreat trend

TRIGGER / MECHANISM

RESCUE AND CHARTING → thin capacity, incomplete surveys
NEAR-TERM TRAFFIC → resource export and destination, not transit container

EFFECT / INTERVENTION

ADVANTAGE → routes shorter than Suez/Panama between NE Asia and NW Europe
MOBILE ICE → unpredictable, hazardous despite retreat trend

DEFINITION / COMPONENTS

- ADVANTAGE → routes shorter than Suez/Panama between NE Asia and NW Europe
- MOBILE ICE → unpredictable, hazardous despite retreat trend

TRIGGER / MECHANISM

- RESCUE AND CHARTING → thin capacity, incomplete surveys
- NEAR-TERM TRAFFIC → resource export and destination, not transit container

EFFECT / INTERVENTION

- ADVANTAGE → routes shorter than Suez/Panama between NE Asia and NW Europe
- MOBILE ICE → unpredictable, hazardous despite retreat trend

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NEAR-TERM TRAFFIC → resource export and destination, not transit container.

08

STAGE 08 INDIA COLD DESERT LADAKH

INDIA COLD DESERT LADAKH GREATER HIMALAYA BLOCKS SW MONSOON, TRANS-HIMALAYA STAYS ARID HIGH ALTITUDE INTENSE INSOLATION BY DAY, SHARP RADIATIVE COOLING BY NIGHT WINTER PRECIPITATION LIGHT SNOW FROM WESTERN DISTURBANCES BIOME CLASS

COLD DESERT, SAME AS SIERRA NEVADA (USA) AND ANDES

RAIN-SHADOW → Greater Himalaya blocks SW monsoon, Trans-Himalaya stays arid

HIGH ALTITUDE → intense insolation by day, sharp radiative cooling by night

WINTER PRECIPITATION → light snow from western disturbances

BIOME CLASS → cold desert, same as Sierra Nevada (USA) and Andes (Argentina)

SYSTEM / DEFINITION

- RAIN-SHADOW → Greater Himalaya blocks SW monsoon, Trans-Himalaya stays arid
- HIGH ALTITUDE → intense insolation by day, sharp radiative cooling by night

PROCESS / MECHANISM

- WINTER PRECIPITATION → light snow from western disturbances
- BIOME CLASS → cold desert, same as Sierra Nevada (USA) and Andes (Argentina)

SPATIAL / INDIA PATTERN

- RAIN-SHADOW → Greater Himalaya blocks SW monsoon, Trans-Himalaya stays arid
- HIGH ALTITUDE → intense insolation by day, sharp radiative cooling by night

HAZARD / LIMIT / RESPONSE

- WINTER PRECIPITATION → light snow from western disturbances
- BIOME CLASS → cold desert, same as Sierra Nevada (USA) and Andes (Argentina)

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BIOME CLASS → cold desert, same as Sierra Nevada (USA) and Andes (Argentina).

09

STAGE 09 INDIA POLAR STATIONS

INDIA POLAR STATIONS DAKSHIN GANGOTRI (1983) FIRST STATION, NOW DECOMMISSIONED/BURIED IN ICE MAITRI (1989) SCHIRMACHER OASIS, ANTARCTICA; OPERATIONAL BHARATI (2012) LARSEMANN HILLS, ANTARCTICA; OPERATIONAL HIMADRI (2008)

DAKSHIN GANGOTRI (1983) → first station, now decommissioned/buried in ice

MAITRI (1989) → Schirmacher Oasis, Antarctica; operational

BHARATI (2012) → Larsemann Hills, Antarctica; operational

HIMADRI (2008) → Ny-Alesund, Svalbard; Arctic; HIMANSH → Spiti, HP

SYSTEM / DEFINITION

- DAKSHIN GANGOTRI (1983) → first station, now decommissioned/buried in ice
- MAITRI (1989) → Schirmacher Oasis, Antarctica; operational

PROCESS / MECHANISM

- BHARATI (2012) → Larsemann Hills, Antarctica; operational
- HIMADRI (2008) → Ny-Alesund, Svalbard; Arctic; HIMANSH → Spiti, HP

SPATIAL / INDIA PATTERN

- DAKSHIN GANGOTRI (1983) → first station, now decommissioned/buried in ice
- MAITRI (1989) → Schirmacher Oasis, Antarctica; operational

HAZARD / LIMIT / RESPONSE

- BHARATI (2012) → Larsemann Hills, Antarctica; operational
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09

STAGE 09 INDIA POLAR STATIONS

INDIA POLAR STATIONS DAKSHIN GANGOTRI (1983) FIRST STATION, NOW DECOMMISSIONED/BURIED IN ICE MAITRI (1989) SCHIRMACHER OASIS, ANTARCTICA; OPERATIONAL BHARATI (2012) LARSEMANN HILLS, ANTARCTICA; OPERATIONAL HIMADRI (2008)

→ DAKSHIN GANGOTRI (1983) → first station, now decommissioned/buried in ice

MAITRI (1989) → Schirmacher Oasis, Antarctica; operational

→ BHARATI (2012) → Larsemann Hills, Antarctica; operational

→ HIMADRI (2008) → Ny-Alesund, Svalbard; Arctic; HIMANSH → Spiti, HP

SYSTEM / DEFINITION

- DAKSHIN GANGOTRI (1983) → first station, now decommissioned/buried in ice
- MAITRI (1989) → Schirmacher Oasis, Antarctica; operational

PROCESS / MECHANISM

- BHARATI (2012) → Larsemann Hills, Antarctica; operational
- HIMADRI (2008) → Ny-Alesund, Svalbard; Arctic; HIMANSH → Spiti, HP

SPATIAL / INDIA PATTERN

- DAKSHIN GANGOTRI (1983) → first station, now decommissioned/buried in ice
- MAITRI (1989) → Schirmacher Oasis, Antarctica; operational

HAZARD / LIMIT / RESPONSE

- BHARATI (2012) → Larsemann Hills, Antarctica; operational
- HIMADRI (2008) → Ny-Alesund, Svalbard; Arctic; HIMANSH → Spiti, HP

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HIMADRI (2008) → Ny-Alesund, Svalbard; Arctic; HIMANSH → Spiti, HP.

10

STAGE 10 INDIA POLAR LEGAL FRAMEWORK

INDIA POLAR LEGAL FRAMEWORK ANTARCTIC TREATY 1959 INDIA ACCEDED 1983; CONSULTATIVE PARTY INDIAN ANTARCTIC ACT 2022 DOMESTIC LEGISLATION FOR POLAR ACTIVITIES ARCTIC POLICY 2022 SIX Pillars; OBSERVER IN ARCTIC FORUM RESEARCH FOCUS

ANTARCTIC TREATY 1959 → India acceded 1983; consultative party

→ INDIAN ANTARCTIC ACT 2022 → domestic legislation for polar activities

→ ARCTIC POLICY 2022 → six pillars; observer in Arctic forum

→ RESEARCH FOCUS → teleconnections: Arctic ice loss to Indian monsoon disruption

SYSTEM / DEFINITION

- ANTARCTIC TREATY 1959 → India acceded 1983; consultative party
- INDIAN ANTARCTIC ACT 2022 → domestic legislation for polar activities

PROCESS / MECHANISM

- ARCTIC POLICY 2022 → six pillars; observer in Arctic forum
- RESEARCH FOCUS → teleconnections: Arctic ice loss to Indian monsoon disruption

SPATIAL / INDIA PATTERN

- ANTARCTIC TREATY 1959 → India acceded 1983; consultative party
- INDIAN ANTARCTIC ACT 2022 → domestic legislation for polar activities

HAZARD / LIMIT / RESPONSE

- ARCTIC POLICY 2022 → six pillars; observer in Arctic forum
- RESEARCH FOCUS → teleconnections: Arctic ice loss to Indian monsoon disruption

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESEARCH FOCUS → teleconnections: Arctic ice loss to Indian monsoon disruption.

11

STAGE 11 POLAR CLIMATE ANSWER SPINE

POLAR CLIMATE ANSWER SPINE COLD DESERT NORTH OF ARCTIC CIRCLE; ICE-CAP VS TUNDRA ARCTIC OCEAN VS ANTARCTIC CONTINENT; FLOATING VS GROUNDED ICE ICE-ALBEDO FEEDBACK, PERMAFROST CARBON, FOUR TRANSMISSION CHANNELS

INDIA'S LADAKH ANALOGUE, POLAR RESEARCH, 2021 PYQ FOUR-CHANNEL SPINE

DEFINE → cold desert north of Arctic Circle; ice-cap vs tundra subdivisions

→ DISTINGUISH → Arctic ocean vs Antarctic continent; floating vs grounded ice

→ EXPLAIN → ice-albedo feedback, permafrost carbon, four transmission channels

→ QUALIFY → India's Ladakh analogue, polar research, 2021 PYQ four-channel spine

DEFINITION / COMPONENTS

- DEFINE → cold desert north of Arctic Circle; ice-cap vs tundra subdivisions
- DISTINGUISH → Arctic ocean vs Antarctic continent; floating vs grounded ice

TRIGGER / MECHANISM

- EXPLAIN → ice-albedo feedback, permafrost carbon, four transmission channels
- QUALIFY → India's Ladakh analogue, polar research, 2021 PYQ four-channel spine

EFFECT / INTERVENTION

- DEFINE → cold desert north of Arctic Circle; ice-cap vs tundra subdivisions
- DISTINGUISH → Arctic ocean vs Antarctic continent; floating vs grounded ice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFY → India's Ladakh analogue, polar research, 2021 PYQ four-channel spine.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT NCPOR (GOA) RUNS INDIA'S POLAR SCIENCE; INDIA ACCEDED TO DAKSHIN GANGOTRI (1ST, 1983–90) MAITRI (1989, SCHIRMACHER OASIS) RESEARCH THEME TELECONNECTIONS — ARCTIC SEA-ICE LOSS IS LINKED TO DISRUPTION

INDIA'S PRINCIPAL COLD-DESERT REGION; DISTINGUISH TRANS-HIMALAYAN KARAKORAM/LADAKH RANGES

COLD-DESERT BIOME SHARED WITH SIERRA NEVADA (USA) AND THE

OPTIONAL PROCESS DEPTH

- CAUTION: NCPOR (Goa) runs India's polar science; India acceded to the Antarctic Treaty (1983) and has
- CAUTION: Stations: Dakshin Gangotri (1st, 1983–90) → Maitri (1989, Schirmacher Oasis) → Bharati
- CAUTION: Research theme: teleconnections — Arctic sea-ice loss is linked to disruption of the Indian
- Ladakh = India's principal cold-desert region; distinguish Trans-Himalayan Karakoram/Ladakh ranges

DATA / INSTITUTION LIMIT

- Cold-desert biome shared with Sierra Nevada (USA) and the Andes (Argentina) — light winter
- CAUTION: India's polar/high-altitude stations: Maitri, Bharati (Antarctic), Himadri (Arctic),
- CAUTION: Legal/policy: Antarctic Treaty 1959 (India 1983), Indian Antarctic Act 2022, Arctic Policy
- NOT: Ladakh is arid because it is hot → arid due to rain-shadow + high altitude (a cold desert).

CONTEMPORARY PARALLEL

- NOT: Bharati is India's Arctic station → Bharati & Maitri are ANTARCTIC; Himadri is the Arctic
- NOT: Dakshin Gangotri is operational → it was the first station (1983), later decommissioned/buried
- NOT: Ladakh gets its rain from the SW monsoon → mainly light winter snow from western disturbances.
- "The Ladakh cold desert is a fragile high-altitude ecosystem under climate and tourism stress." Discuss

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.